Technical Notes

Sample. Starting Cohort 2 (SC2) was recruited in kindergarten (kindergarten sample) and supplemented by an augmentation sample in Grade 1. I excluded the kindergarten waves. SC2 therefore covers Grades 1 to 4 (2013 to 2016). I also excluded 2,420 children who moved to a school that is not part of the NEPS sample, because NEPS does not conduct standardised testing at these schools. SC3 started in Grade 5 and covers Grades 5 to 12 (2010 to 2018). For SC3, I excluded 845 participants from Brandenburg and Berlin, where students switch to secondary school after Grade 6. NEPS classifies them as school changers and does not test them. The analytic sample comprises 14,355 individuals: 6,892 in SC2 and 7,463 in SC3.

Weights. All models used the NEPS cross-sectional weights of the wave in which students entered the sample, rescaled to a mean of one within each cohort. The longitudinal weights reduced the sample to N = 7,119. The weights account for unequal selection probabilities and nonresponse at entry.

Missing data. Item nonresponse and panel attrition were handled by multiple imputation by chained equations with predictive mean matching (five nearest neighbours) in Stata 19, assuming data are missing at random conditional on the imputation model (Van Buuren, 2018). Imputation was carried out separately for SC2 and SC3, so that grades not covered by a cohort by design were not imputed. The imputation model included all analysis variables, federal state, panel status and survey wave; in SC3 it also included school track. Auxiliary variables were selected based on theoretical relevance and a correlation of $r > .3$ with an analysis variable (Graham, 2012). For SC2, these were household income, books at home, language spoken at home, mathematics grade, and migrant-origin class and school composition. For SC3, school size and German grade were added. I used 20 burn-in cycles between imputations. A pilot imputation yielded FMI values of up to 0.95. Following (White et al., 2011), I therefore generated 100 imputed datasets.

References

Graham, J. W. (2012). *Missing data: Analysis and design*. Springer Science & Business Media. https://doi.org/10.1007/978-1-4614-4018-5

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White, I. R., Royston, P., & Wood, A. M. (2011). Multiple imputation using chained equations: Issues and guidance for practice. *Statistics in Medicine*, *30*(4), 377-399. https://doi.org/10.1002/sim.4067